

Customer Dynamic Engagement Analysis

Time-Series Activity, Statistical Thresholds and K-Means Clustering

Project Area	Women's Clothing Trends and Analysis in the Fashion Industry
Dataset	E-Commerce Customer Behavior and Sales Dataset
Notebook	Customer_Dynamic_Engagement_Analysis.ipynb
Data	5,000 transactions; January 2023 to March 2024

1. Objective

Generate a feature that identifies highly engaged customers by analysing chronological session behaviour, applying a statistical cutoff and clustering customers according to their engagement levels. The final feature is Highly_Engaged, where 1 means highly engaged and 0 means not highly engaged.

2. Dataset Suitability

The dataset provides Customer_ID, Date, Order_ID, Product_Category, Session_Duration_Minutes, Pages_Viewed and Total_Amount. These fields support time ordering, session analysis, customer aggregation and engagement scoring.

Dataset limitation: Product_Category contains a general Fashion label but does not separate women's, men's and unisex clothing. Fashion transactions are therefore used as a proxy, and no unsupported women's-clothing-only claim is made.

3. Features Used

Feature	Purpose
Session_Duration_Minutes	Measures time spent on the website
Pages_Viewed	Measures browsing depth
Total_Amount	Represents purchase value
Order_ID / Customer_ID	Measures customer frequency and history

4. Required Methodology

- Convert Date to datetime, remove duplicates and sort each customer's transactions chronologically.
- Filter Product_Category = Fashion to align the records with the project area.
- Calculate rolling three-session averages for duration, pages viewed and purchase amount.
- Convert rolling values to percentile ranks and calculate a dynamic score from 0 to 100.
- Aggregate customer behaviour, standardize the values and apply K-Means with three clusters.
- Use the 75th percentile as the statistical high-engagement threshold.
- Set Highly_Engaged = 1 only for customers above the threshold and in the strongest cluster.

4.1 Dynamic Engagement Score

Score = 100 x (0.35 Session + 0.35 Pages + 0.20 Purchase + 0.10 Frequency)

Component	Weight	Meaning
Rolling session duration	35%	Time spent interacting
Rolling pages viewed	35%	Browsing depth
Rolling amount	20%	Purchase value
Session number	10%	Repeated activity

4.2 Core Implementation

```
df["Date"] = pd.to_datetime(df["Date"])
df = df.drop_duplicates()
df = df[df["Product_Category"].str.lower().eq("fashion")]
df = df.sort_values(["Customer_ID", "Date"])

df["Rolling_Session"] = df.groupby("Customer_ID")["Session_Duration_Minutes"]
    .transform(lambda x: x.rolling(3, min_periods=1).mean())

df["Rolling_Pages"] = df.groupby("Customer_ID")["Pages_Viewed"]
    .transform(lambda x: x.rolling(3, min_periods=1).mean())

df["Session_Score"] = df["Rolling_Session"].rank(pct=True)
df["Page_Score"] = df["Rolling_Pages"].rank(pct=True)
df["Dynamic_Engagement_Score"] = 100 * (
    0.50 * df["Session_Score"] + 0.50 * df["Page_Score"]
)
```

5. Clustering and Final Feature

```
features = ["Average_Session_Duration", "Average_Pages_Viewed",
            "Purchase_Frequency", "Total_Spending",
            "Average_Engagement_Score"]

scaled_data = StandardScaler().fit_transform(customer_engagement[features])
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
customer_engagement["Cluster"] = kmeans.fit_predict(scaled_data)

threshold = customer_engagement[
    "Average_Engagement_Score"
].quantile(0.75)

customer_engagement["Highly_Engaged"] = np.where(
    (customer_engagement["Average_Engagement_Score"] >= threshold)
    & (customer_engagement["Engagement_Level"] == "High Engagement"),
    1, 0
)
```

6. Expected Output

The process adds Dynamic_Engagement_Score, Cluster, Engagement_Level and Highly_Engaged. Cluster numbers are not interpreted directly; clusters must first be ordered by their average engagement score and named Low, Medium and High.

Field	Interpretation
Dynamic_Engagement_Score	Higher score means stronger recent engagement
Engagement_Level	Low, Medium or High customer group
Highly_Engaged	1 = high cluster and above 75th percentile; otherwise 0

7. Conclusion

This task combines time-ordered session patterns, percentile-based scoring, a statistical threshold and K-Means clustering to create an interpretable Highly_Engaged feature. The exact threshold and customer counts must be obtained after running the notebook on the actual CSV; no numerical results are assumed in this report.

Files generated after execution

- fashion_dynamic_engagement_results.csv
- customer_engagement_summary.csv